

This is our major newspaper source.

It provides several useful statistics:

- more than **120,000 fishers and fish farmers** whose livelihoods are connected to fisheries;
- demersal fish biomass/density down **88%** compared with virgin-stock levels in the 1960s;
- Malaysians consumed approximately **53 kg of seafood per capita in 2020**, compared with about 40 kg for Southeast Asia and 20 kg globally;
- discusses overfishing, habitat destruction and high seafood consumption;
- explicitly advocates responsible seafood consumption and avoiding seafood on WWF's "Avoid" list.



Saving our seafood, from boat to bowl

MALAYSIA, a nation blessed with rich marine biodiversity, has long relied on its fisheries sector as a...



Author CHITRA DEVI

2. Malaysian consumer research

"Fish Purchasing Behavior and Implications for Promoting Sustainable Consumption: A Malaysian Case" — 2023

This is probably the **most directly relevant academic source for our user/problem definition**.

The study surveyed **250 Malaysian consumers** and found that interventions should address:

- low health and sustainability literacy;
- limited availability of sustainable options;
- limited accessibility;
- affordability barriers.

It also specifically discusses sustainable fish purchasing behaviour and interventions in Malaysia.

DOI:

10.13106/ajbe.2023.vol13.no2.29



Fish Purchasing Behavior and Implications for Promoting Sustainable Consumption: A Malaysi...
Asian Journal of Business Environment, 2023, 13(2), 29



https://www.kci.go.kr/kciportal/landing/article.kci?arti_id=ART002954995&utm_source=chatgpt.com

DOI / article — 10.13106/ajbe.2023.vol13.no2.29

3. The 73% willingness-to-pay evidence

"Consumers' Awareness, Perception and Willingness-to-Pay for MyGAP-Compliant Farmed Fish Produce: Evidence from Malaysia" — 2025

This is the source behind our **73% statistic**.

The study found that **over 73% of surveyed consumers expressed willingness to pay a premium for MyGAP-compliant farmed fish**. It also found that aquaculture knowledge, green consumer behaviour and MyGAP awareness influence willingness to pay.

Important wording:

▮ **"Over 73% of surveyed consumers..."**

Not:

▮ "73% of Malaysians..."

The full open-access paper is available here.



Consumers' Awareness, Perception and Willingness-to-Pay for MyGAP-Compliant Farmed Fish...
Aquaculture is essential for the global fish supply, but sustainable practices in Malaysia face significant challen...

https://journals.nasspublishing.com/index.php/rwae/article/view/2192?utm_source=chatgpt.com

Nasspublishing

https://journals.nasspublishing.com/index.php/rwae/article/download/2192/1139?utm_source=chatgpt.com

DOI — 10.36956/rwae.v6i3.2192

4. Demand-side intervention research

These are the sources supporting our argument that **consumer demand can be a legitimate intervention point** rather than SukaSeafood simply being an educational app.

"Information, altruism, and the value of traceability in seafood" — Fisheries Research, 2026

This is the most important theoretical source for our demand-side framing.

The paper explicitly states that one potential approach to addressing unsustainable seafood practices is **demand-side intervention through information provision**. It investigates consumer responses to information related to seafood traceability, resource conservation and other attributes.

ScienceDirect

https://www.sciencedirect.com/science/article/pii/S0165783626000998?utm_source=chatgpt.com



https://www.sciencedirect.com/science/article/pii/S0165783626000998?utm_source=chatgpt.com

"Can demand-side management replicate a size limit in a small-scale fishery?" — Fisheries Research

This is an earlier, useful proof-of-concept for the broader principle that **economic/market incentives can influence fishing behaviour**.

In the Indonesian small-scale fishery experiment, price premiums shifted the size distribution of catches; one species showed a 15% increase in average length and overall average fish size increased by 8%.

It doesn't directly prove that an app like SukaSeafood will change Malaysian fisheries, so use it as **supporting evidence for market/demand mechanisms**, not as a direct analogue.

ScienceDirect

https://www.sciencedirect.com/science/article/abs/pii/S0165783619302917?utm_source=chatgpt.com

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https://www.sciencedirect.com/science/article/abs/pii/S0165783619302917?utm_source=chatgpt.com

Nature Food — "A meta-analysis assessing the effectiveness of demand-side interventions for sustainable food consumption and food waste reduction" — 2026

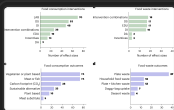
This is our broader behavioural-science evidence.

The meta-analysis covered **306 effect sizes from 110 articles and more than 2.4 million observations**. It found generally small average effects, but meaningful variation by intervention type, with some choice-architecture interventions showing stronger effects.

This is useful because it lets us make a **careful** claim:

▮ Demand-side interventions can influence behaviour, but intervention design matters.

It does **not** justify claiming that SukaSeafood will definitely change behaviour.



A meta-analysis assessing the effectiveness of demand-side interventions for sustainab...
Nature Food – Demand-side food policies show small average effects, with effectiveness varying across...

n

https://www.nature.com/articles/s43016-025-01279-9?utm_source=chatgpt.com

5. SDG 14 official sources

United Nations — SDG 14

This is the authoritative source for the actual SDG targets.

Relevant targets:

14.2

Sustainably manage and protect marine and coastal ecosystems.

14.4

Effectively regulate harvesting, end overfishing and destructive fishing practices, and restore fish stocks.

14.5

Conserve coastal and marine areas.

14.a

Increase scientific knowledge and develop research capacity and marine technology.



Goal 14 | Department of Economic and Social Affairs

Conserve and sustainably use the oceans, seas and marine resources for sustainable development

https://sdgs.un.org/goals/goal14?utm_source=chatgpt.com

This is the source we should use whenever we state what **14.2 / 14.4 / 14.5 / 14.a** actually mean.

6. Core product datasets

These are the sources underpinning the actual SukaSeafood system.

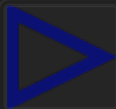
WWF Save Our Seafood

Used for:

- Seafood Discovery
- Sustainability Intelligence
- future Smart Swap
- comparison

Key information:

- species/common names;
- sustainability classification;
- origin;
- production method;
- certification/notes;
- alternatives.



Home - Save Our Seafood

Search our recommendations Species Refine by: BEST CHOICE REDUCE AVOID Browse Save Our Seafood G...

 https://www.saveourseafood.my/?utm_source=chatgpt.com

7. OpenDOSM PriceCatcher

This is the foundation of **Price Intelligence**.

The current OpenDOSM documentation says PriceCatcher contains **more than one million price records per month**, with prices collected and verified by ground staff daily. The transactional dataset contains:

- date;
- premise code;
- item code;
- price.

The premise lookup adds:

- premise;
- address;
- premise type;
- state;
- district.

The item lookup adds:

- item name;
- unit;
- item group;
- item category.

Very importantly, OpenDOSM says PriceCatcher is designed for **high-frequency analysis of specific items in specific locations**, not as a nationally representative CPI dataset. That is why we've deliberately called the UI metric **observed price context** rather than "the Malaysian average price."

Main dataset



PriceCatcher: Transactional Records | OpenDOSM

The table below provides a preview of the full dataset, which contains over a million price records p...



Author Department of Statistics Malaysia

Item Lookup



PriceCatcher: Item Lookup | OpenDOSM

Lookup table, to be left-joined against item code in the main PriceCatcher data.



Author Department of Statistics Malaysia

Premise Lookup



PriceCatcher: Premise Lookup | OpenDOSM

Lookup table, to be left-joined against premise code in the main PriceCatcher data.



Author Department of Statistics Malaysia

8. MyBIS — Malaysia Biodiversity Information System

MyBIS is the Malaysian Biodiversity Information System and describes itself as a one-stop repository for biodiversity information in Malaysia.

It supports exploration by:

- species;
- specimens;
- photographs;
- taxonomy;
- datasets;
- maps;
- protected areas;
- references/literature.

This is our potential future foundation for:

- species-linked ecosystem context;
- Malaysian biodiversity;
- protected-area context.



Malaysia Biodiversity Information System

Explore Malaysia's biodiversity through the MyBIS platform, providing access to flora, fauna, ecosystems, an...



https://www.mybis.gov.my/?utm_source=chatgpt.com

MyBIS protected areas

MyBIS has a dedicated protected-area explorer with Malaysian protected-area records.



Malaysia Biodiversity Information System

The primary objective of this portal is to provide a mechanism for the efficient exchange of information on bi...



https://www.mybis.gov.my/one/pamaps.php?type=8&utm_source=chatgpt.com

9. OBIS — Ocean Biodiversity Information System

OBIS provides open marine biodiversity occurrence data from thousands of datasets in an integrated system. It supports visual exploration as well as programmatic access through APIs, downloadable datasets and AWS Open Data.

Relevant fields include occurrence/taxonomic information and the data provided by the source datasets.

This is the basis for the possible future:

Marine Biodiversity Context

Access OBIS data | Ocean Biodiversity Information System (OBIS)

OBIS harvests occurrence records from thousands of datasets and makes them available as a single integrated...

https://portal.obis.org/data/access/?utm_source=chatgpt.com

OBIS Data policy | Ocean Biodiversity Information System (OBIS)

This data policy is an extract from the OBIS Manual. Please refer to the manual for the latest version of the OBI...

https://www.obis.org/data/datapolicy/?utm_source=chatgpt.com

10. Public fish-image datasets

For the **future Computer Vision epic**, our concept has been based on using public labelled fish-image datasets.

We deliberately did **not** lock ourselves to one dataset in the final discovery presentation because the supported Malaysian species set needs to be determined by the intersection of:

Malaysian relevance + available labelled images + reliable species mapping.

So I would **not present a specific Kaggle dataset as a committed source yet** unless the team has independently validated it.

This is an important distinction from our earlier ideation, where we discussed Fish-4K / large fish-image datasets.

11. Design-thinking artefacts

These aren't external sources, but they are the design artefacts we've developed from the evidence above.

Problem statement

WHO: Everyday Malaysian seafood consumers

WHAT: Lack decision support connecting seafood identity, sustainability and price

WHERE: Point of purchase

WHY: Information is fragmented and disconnected from the actual decision

Persona

Amir Rahman, 29

Malay Malaysian

Kuala Lumpur

Busy professional

Weekly seafood shopper

Moderately sustainability-conscious

Not an environmental activist

Empathy map

THINK

- Is this fish becoming overfished?
- Is this actually sustainable?
- Is there a better alternative?

FEEL

- Curious
- Uncertain
- Concerned about Malaysia's environment
- Willing to make a better choice if it is easy

SAY

- "I've heard fish stocks are declining."
- "I usually just buy what I know."
- "Why can't I find this fish as often as I used to?"

DO

- Checks price
- Buys familiar seafood

- Looks at appearance
- Rarely researches sustainability before shopping

User journey

Enter store → See seafood → Identify → Understand → Check price → Decide → Purchase

The intervention point is the gap between:

| UNDERSTAND

and

| DECIDE

Theory of change

Data → Information → Understanding → Better choice → Repeated choice → Potential demand shift

Product journey

IDENTIFY → UNDERSTAND → COMPARE → ACT → REPEAT

12. The four current Iteration 1 product artefacts

These are the UI renders we generated and used in the discovery deck:

Search UI

Supports **Seafood Discovery**

Seafood Profile UI

Supports **Seafood Discovery + Sustainability Intelligence**

Price Intelligence UI

Supports **Price Intelligence**

Home / Main Product UI

Shows the broader product concept and future-facing product surface.

These aren't external research sources; they are our **prototype artefacts**.

13. Important claims we've deliberately avoided

These are just as important as the sources because we've actively corrected several potential hallucinations during development.

We should **not** claim:

- Malaysia is the world's largest seafood market;
- 73% of all Malaysians will pay more for sustainable seafood;
- SukaSeafood knows where an individual fish was caught;
- SukaSeafood can verify fisher provenance;
- PriceCatcher is a guaranteed national average;
- OBIS proves a species is currently present at a location;
- WWF provides a public API unless we verify one;
- SukaSeafood directly restores fish stocks;
- consumer education by itself guarantees SDG 14 impact.

These limitations are part of what makes the project defensible.

14. The essential source pack

1. [WWF Malaysia — Food / Sustainable Seafood](#)
2. [WWF Save Our Seafood](#)
3. [The Star — Saving our seafood, from boat to bowl](#)
4. Fish Purchasing Behavior — Malaysian Case
5. MyGAP Consumer WTP Study — 2025
6. [Fisheries Research — Information, altruism and seafood traceability](#)
7. [Nature Food — Demand-side interventions meta-analysis](#)
8. [UN SDG 14 — Life Below Water](#)

The **technical data pack**:

1. [OpenDOSM PriceCatcher](#)
2. [OpenDOSM Item Lookup](#)
3. [OpenDOSM Premise Lookup](#)
4. [MyBIS](#)
5. [OBIS Data Access](#)